Automotive Cyber Security Academy 2023

Basic Information Technology Security

Topic 2: Terminology and concepts for IT security

Automotive Cyber Security Academy

1 2 9 0 FACULDADE DE CIÊNCIAS E TECNOLOGIA UNIVERSIDADE D COIMBRA

dei departamento de engenharia informática

30/04/23

Outline

- Fundamental Concepts
  - CIA triad
- Security Attacks
  - Passive
  - Active
  - DarkWeb
- Type of Attackers & Their Motivativations
  - Black Hat
  - Gray Hat
  - White Hat

Fundamental Concepts

Fundamental Concepts - Confidentiality

- Confidentiality
  - Data Confidentiality – Mechanisms to assure that private or confidential information is not disclosed to unauthorized persons
  - Privacy – Assures that individuals control how information related to them can be collected, processed, stored and whom can access the information
- Can be supported through encryption techniques.

Fundamental Concepts - Integrity

- Integrity
  - Data Integrity – Mechanisms to assure that information is only modified in a specific and authorized manner
  - System Integrity – Assures that a system performs the function it was designed for, that is not disrupted from unauthorized manipulation

- Can be supported with Secure Hashing approaches (HMAC)

Fundamental Concepts - Availability

- Availability
  - Assures that systems provide their functions in a promptly mode, without denying functionalities to authorized users

- Can be supported with authentication, authorization services and with replication techniques.

Fundamental Concepts

Fundamental Concepts – Authenticity

- Authenticity
  - Assures that information can be verified can be trusted, confidence in the validity of a transmission, a message or the message source
  - Mechanisms to verify that users are **Who** they claim to be, and that inputs come from a trusted source.

- Can be supported with mutual authentication and asymmetric encryption.

Fundamental Concepts – Accountability & Non-Repudiation

- Accountability
  - Assures that actions of an entity can be traced uniquely to that entity.
- Non-Repudiation
  - Assures that no party can deny that it sent or received a message
- Can be supported with encryption and digital signatures

Fundamental Concepts – Vulnerability, Threat, Attack

- As per IETF RFC 4949 “Internet Security Glossary, Version 2”
- Vulnerability
  - A flaw or weakness in a system’s design, implementation of operation that can be exploited
- Threat
  - Potential violation of security that exists when there in an event that can breach security and cause harm
- Attack
  - Violation of security policies of a system, evasion of security services

Security Attacks

Security Attacks - as per Intent

- Passive
  - Tries to learn of make use of information from the system but does not affect the system
  - Useful for an off-line analysis
- Active
  - Tries to affect the system resources of affect the operation of system (availability)

Security Attacks - as per Point of Initiation

- Inside
  - Attack that is initiation by an entity inside the security perimeter, from an insider person in an organization
- Outside
  - Attack initiated from the outside of the security perimeter by an unauthorized or illegitimate user of the system.

Security Attacks - as per Method of Delivery

- Direct
  - Attack is directly delivered to the system, packets are delivered to the intended victim(s)
- Indirect
  - Third parties are used to conduct an attack, to a victim. This approach is used in amplification attacks like Denial of Service (DoS)

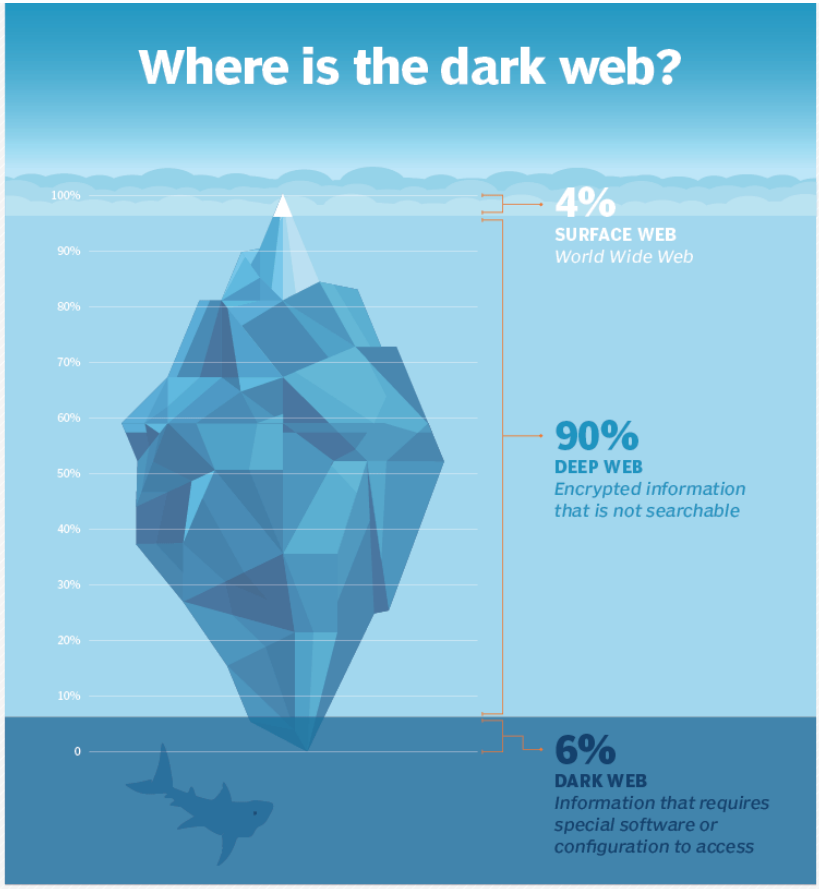
Passive Security Attacks – as per Intent

- Eavesdropping transmissions (listening without authorization)
- Monitoring transmissions (who is communication with whom)
- The goal of the attack is to retrieve the information that is being transmitted
- Difficult to detect, efforts should be on prevention
- Examples:
  - Release message contents (when containing sensitive information)
  - Traffic Analysis

Active Security Attacks – as per Intent

- Involve the modification of the data stream or creation of a false data stream
- In a security perspective the goal is to detect attacks and to recover from any disruption
- Difficult to prevent due to the wide variety of potential vulnerabilities
- Examples:
  - Masquerade – when one entity pretends to be another entity
  - Replay – requires capture of data and subsequent retransmission
  - Modification of Messages – Modify legitimate messages
  - Denial of Service

Dark Web, darknet, deepnet



- Encrypted part of the Internet that is not indexed
- Requires specific configuration or authorization access
- Used by criminal elements but also by benign persons that seek for privacy
- Information exchange in a Peer-to-Peer network connection or overlay connection like Tor.
- Mainly used to buy and sell stolen data.

Source [TechTarget]

Type of Attackers/Hackers

Types of Attackers/Hackers

Source TechTarget

- Black Hat – With Malicious intents, theft of information, fraud or service disruption
- White Hat – Cybersecurity experts who conduct penetration tests, vulnerability assessment
- Gray Hat – with some ethical indecisions, can use illegal methods to find flaws and then expose vulnerabilities

Reflection

Reflections

- Which hacker groups do you know ?
- What have they done ?
- Any interaction with Dark Web ? Should we use it?

Thanks

Automotive Cyber Security Academy

The NEW and IMPROVED - ACSA 2023 is here!